

Ex No:1	PASSPORT AUTOMATION SYSTEM
Date:	

Aim

To draw the UML diagrams namely Use Case, Class, Activity, Sequence, Collaboration, State Chart, Component, Deployment and Package diagrams for the Passport Automation System using Umbrello UML Modeller.

Software Requirements Specification

1.0 Hardware Requirements

- Intel Core i3/i5 Processor
- 4 GB RAM or above
- 500 GB Hard Disk
- Windows 10/11

1.1 Software Requirements

- Umbrello UML Modeller
- Linux/Windows Operating System

1.2 Problem Analysis and Project Plan

The Passport Automation System is developed to simplify the passport application process. The applicant registers in the system, submits the required documents, and the application is verified by the passport officer and police department. After successful verification, the passport is approved and issued to the applicant.

1.3 Project Description

The Passport Automation System provides an online platform for passport registration and verification. It stores applicant information securely, verifies submitted documents, performs

police verification, and enables passport officers to approve or reject applications. The system reduces paperwork, improves efficiency, and speeds up passport issuance.

1.4 Reference

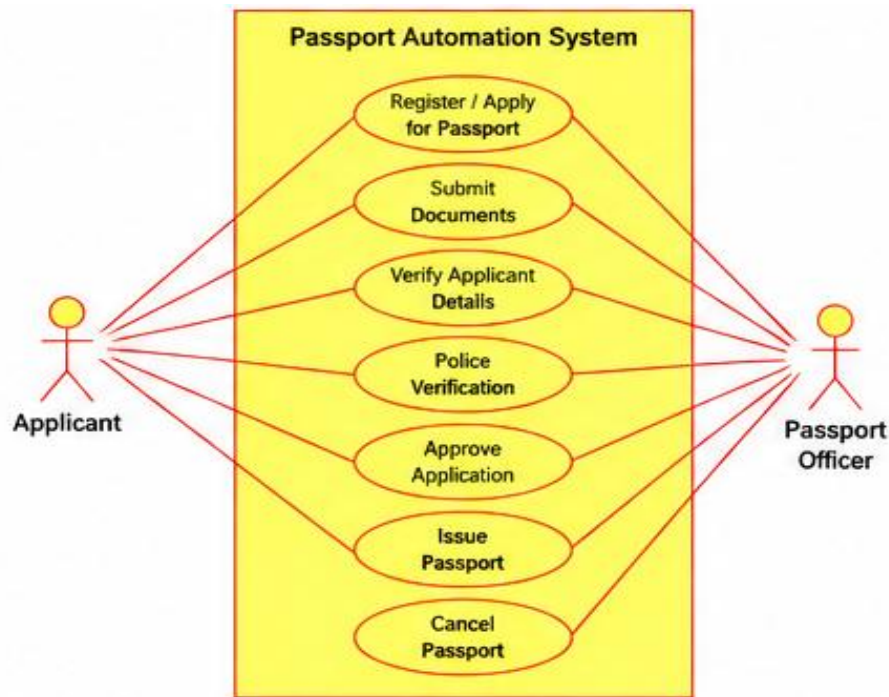
- IEEE Software Requirement Specification (SRS) Format.
- UML 2.5 Standard Documentation.
- Umbrello UML Modeller Documentation.

Procedure

1. Open **Umbrello UML Modeller**.
2. Create a new UML project.
3. Select the **Use Case Diagram** and create actors and use cases.
4. Draw the **Class Diagram** with classes, attributes, methods, and relationships.
5. Create the **Activity Diagram** to represent the workflow.
6. Draw the **Sequence Diagram** showing the interaction between objects.
7. Create the **Collaboration (Communication) Diagram**.
8. Draw the **State Chart Diagram** representing different system states.
9. Create the **Component Diagram** showing software components.
10. (If required) Draw the **Deployment Diagram**.
11. (If required) Draw the **Package Diagram**.
12. Save the project and verify all diagrams.

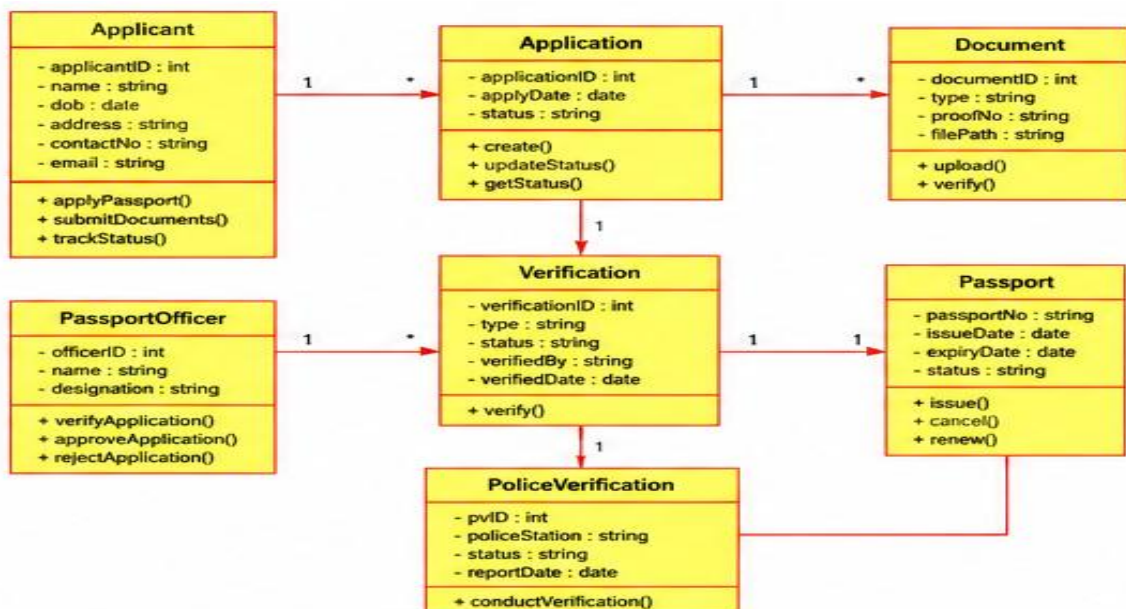
Use Case Diagram

Description:



- **Actors:** Applicant, Passport Officer
- **Use Cases:** Register Application, Submit Documents, Verify Applicant Details, Police Verification, Approve Application, Issue Passport, Cancel Passport.

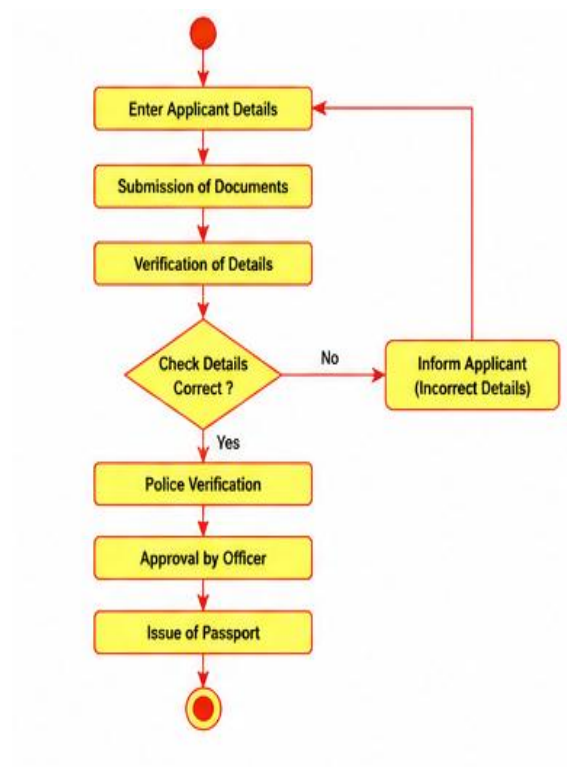
Class Diagram



Description:

- **Classes:** Applicant, Application, Document, Verification, Passport Officer, Passport.
- **Relationships:** Association and dependency between the classes.

Activity Diagram

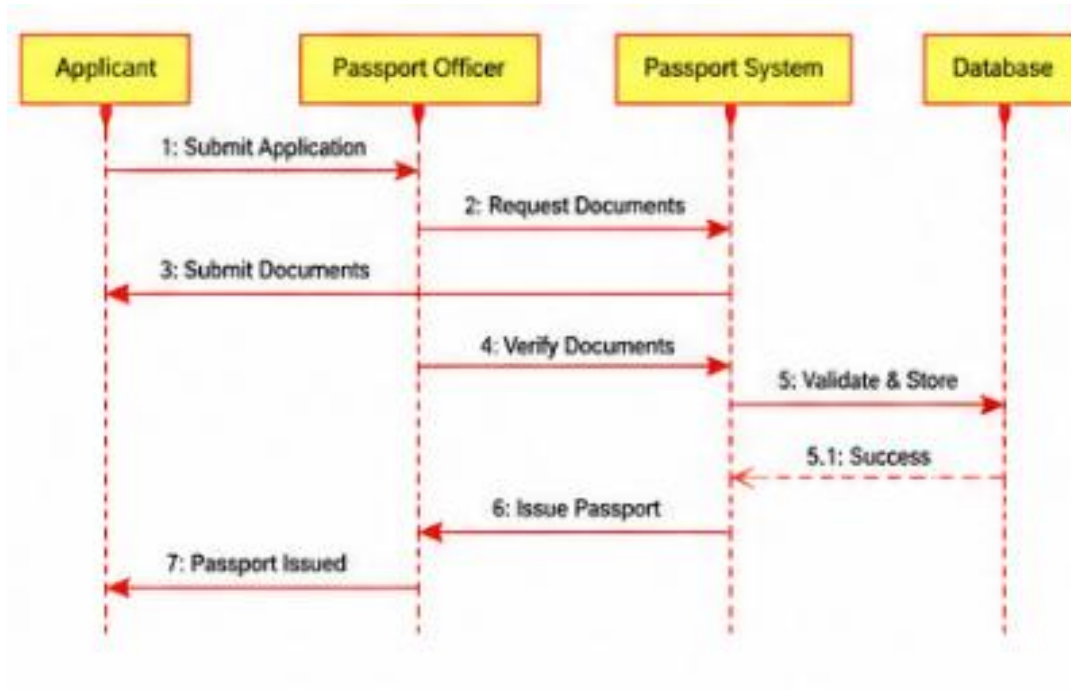


Description:

- Shows the complete workflow from application submission to passport issuance.

Sequence Diagram

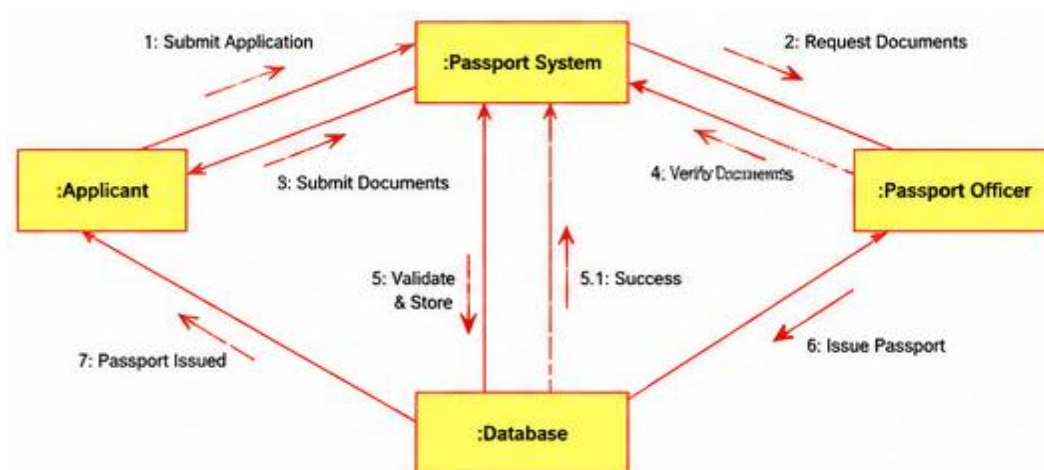
ITEM	DETAILS
OBJECTS	Applicant, passport officer passport system,
START	Applicant submits application
Process	Documents are verified
END	Passport is issued



Description:

- Shows the interaction between Applicant, Passport Officer, Passport System, and Database in chronological order.

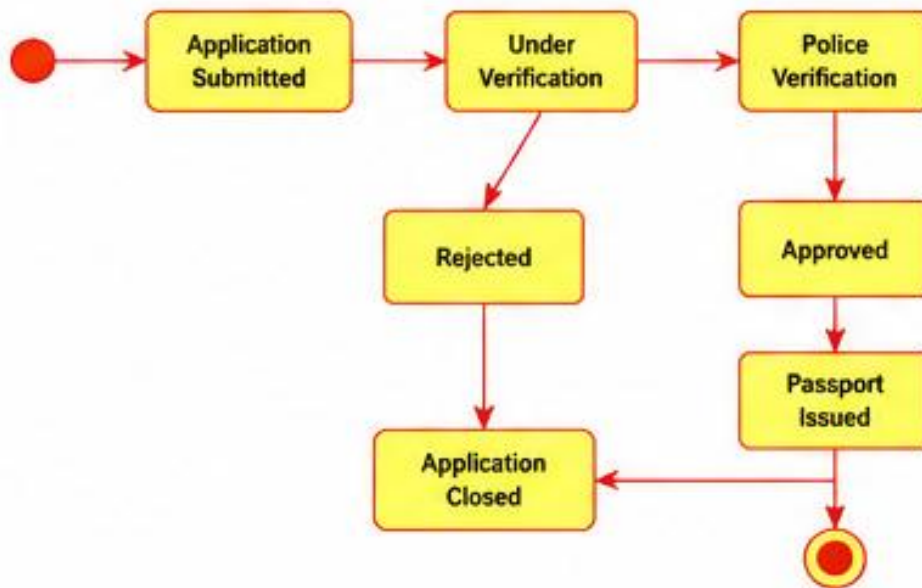
Collaboration Diagram



Description:

- Represents communication among system objects during passport processing.

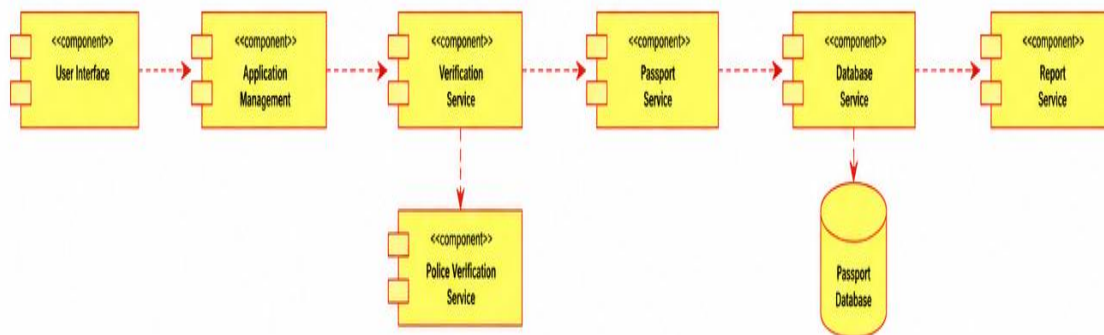
State Chart Diagram



Description:

- Represents different states of a passport application such as Submitted, Under Verification, Approved, Rejected, and Issued.

Component Diagram



Description:

- Represents the software components such as User Interface, Application Management, Verification Service, Passport Service, Database Service, and Report Service.

Result

Thus, the UML diagrams namely **Use Case, Class, Activity, Sequence, Collaboration, State Chart, Component, Deployment, and Package diagrams** for the **Passport Automation System** were successfully designed using Umbrello UML Modeller and verified